LatentWire: No-Text Candidate-Transfer Packets under Destructive Controls

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Abstract

Multi-model language systems usually communicate by emitting text or by sharing dense state such as KV caches. We study a narrower interface: can a source endpoint send a byte-scale packet that helps a public candidate decoder on multiple-choice tasks without sending source text, hidden states, KV cache, logits, score vectors, or labels? We present LatentWire, a no-text packet protocol and destructive-control framework with explicit byte accounting. On ARC-Challenge, an 8-byte payload / 11-byte framed packet reaches 0.344 accuracy across 10 seeds, above target-only 0.265 and same-budget structured text 0.300. On OpenBookQA, a 3-byte / 6-byte packet reaches 0.378 across 20 seeds, above target-only 0.276 but not above same-budget text under the lower-bound gate. The decisive audit is limiting: explicit source-index communication reaches 0.346 on ARC-Challenge and 0.378 on OpenBookQA; an aggregate packet-minus-source-index bootstrap is negative on ARC-Challenge and effectively tied on OpenBookQA. A validation-only 3-byte source-score sketch also ties, rather than beats, the source top-label rule. We therefore claim byte-scale same-family candidate transfer under destructive controls, not evidence synthesis beyond the source, formal privacy, solved latent communication, C2C superiority, or native GPU latency/HBM/energy wins.

1 Introduction

Future language-model systems will often use more than one model: a cheap model may screen candidates, a specialist may inspect hard cases, and a stronger receiver may combine evidence. The simplest interface is text. Recent cache-level systems instead communicate through model internals, such as fused or selectively shared KV caches (Fu et al., 2026; Shi et al., 2026; Ye et al., 2025). These approaches are powerful, but they couple communication to large hidden-state objects and serving substrates.

This paper asks whether a much smaller interface can carry useful source-side candidate evidence. In our setting, the source endpoint and receiver/decoder see the same public question and answer choices. The source performs private computation over those choices. The only communicated object is a short byte packet; the receiver/decoder never sees source text, source logits, source hidden vectors, KV cache, score vectors, or the answer key. The headline receiver in this paper is not a live target-model forward pass. It is a public hashed/DCT candidate decoder built from public task features and frozen receiver-side candidate surfaces; "target-only" means the same decoder with an empty source packet. This is a side-information problem in the spirit of distributed source coding (Slepian & Wolf, 1973; Wyner & Ziv, 1976; Whang et al., 2021; Özyılkan et al., 2023): the receiver already has the public prompt and its own side information, so the packet should carry only the source evidence that is useful at the decoder. We use that framing as motivation, not as a formal rate-distortion theorem: this paper does not specify a full joint distribution, distortion measure, or converse. The scientific contribution is therefore not just a packet row, but a controlled protocol for asking whether such rows survive the controls that usually create false positives.

Our strongest current result is deliberately scoped. On ARC-Challenge and OpenBookQA (Clark et al., 2018; Mihaylov et al., 2018), same-family Qwen2.5 source-cache-to-public-decoder packet transfer (Yang et al., 2025) improves over target-only. ARC-Challenge also clears the same-budget text gate, while OpenBookQA is statistically indistinguishable from same-budget text under the paired-bootstrap lower-bound gate. A strict Phi-3 source-family replacement (Abdin et al., 2024) fails, and additional connector repairs fail on the available TinyLlama/Qwen cached artifacts. This boundary matters: the paper is not claiming solved cross-family latent communication.

The contributions are:

- A byte-scale no-source-text/source-state-private packet protocol and threat model for source-model-to-public-decoder multiple-choice candidate transfer with no source text, hidden vector, KV cache, source-logit, score-vector, or gold-label exposure.
- A strict evaluation suite, drawing on control-task and behavioral-testing methodology (Hewitt & Liang, 2019; Adebayo et al., 2018; McCoy et al., 2019; Ribeiro et al., 2020), that combines target-only, same-budget text, source-index/rank, coordinate-shuffle, wrong-row, same-source-choice, cross-family, and cached-connector controls.
- Controlled narrow positives on ARC-Challenge and OpenBookQA with seed repeats and paired uncertainty, accompanied by an explicit source-index audit showing that the current effect is mostly source-choice preserving.
- A failure-boundary analysis showing that Phi-3 cross-family transfer fails because the packet faithfully transports a weaker alternate source choice, not because the decoder corrupts the packet; simple cached connector repairs also fail, so broad latent-language and cross-family claims are unsupported.
- A systems accounting artifact comparing raw, framed, cacheline-rounded packet sizes with dense KV/cache byte regimes while explicitly separating shape-restricted task hints from general-purpose state-transfer methods.

This is useful even with a narrow positive because it separates three questions that are often conflated: whether a compact private packet can help, whether it carries more than the source's preferred answer, and whether it is cheaper or less exposing than dense state transfer. The current answer is positive for the first question, negative for the second, and accounting-only for the third. That scoped result is the paper's intended workshop contribution.

Where the current result is useful. The current method is most plausible for same-family cascades and bandwidth-limited candidate reranking where sending source generations, logits, hidden states, or KV cache is undesirable. It is not yet a method for open-ended generation, heterogeneous cross-family collaboration, or reasoning synthesis beyond the source model's selected candidate.

2 No-text packet transfer

Let a public multiple-choice instance be $x = (q, c_1, \ldots, c_m)$ with answer choices c_i. A source model S and receiver/decoder R both observe x, but only S computes the source-side private evidence $e_S(x)$. A packet encoder emits

$$z = E(e_S(x), x), \qquad |z| \leq B \text{ bytes}.$$

The receiver predicts

$$\hat{y} = D_R(x, z),$$

where D_R may use public task features and the receiver's own state, but may not access source text, source hidden states, source KV cache, source logits, or answer labels. In the reported headline rows, R is instantiated as a public hashed/DCT candidate decoder rather

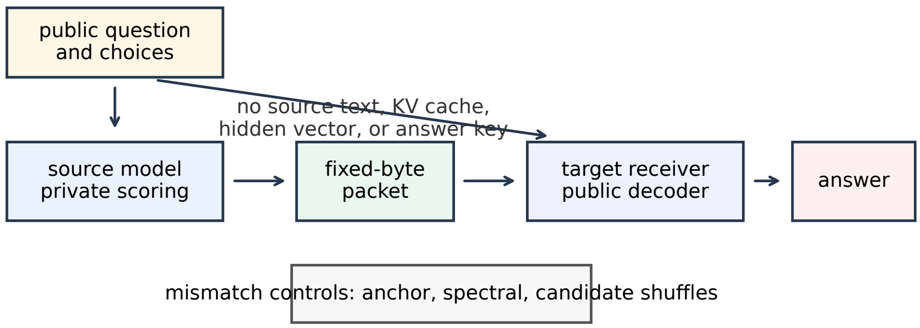

Figure 1: LatentWire protocol. Both sides receive the public task. The source privately scores candidates and emits only a fixed-byte packet. The public receiver/decoder decodes the packet with a public basis and its own side information. Destructive controls scramble the shared coordinate system or candidate relation.

than a live LM endpoint. The target-only row is $D_R(x, \emptyset)$ on the same public decoder with no source packet.

We evaluate a packet only if it beats three surfaces: target-only $D_R(x, \emptyset)$, a same-budget structured text control, and destructive controls that preserve superficial packet size but destroy the intended source-receiver relation. We also audit explicit source-index, source-label-text, source-rank-code, source-score-style, and entropy-matched random-index baselines. The same-budget text control is not the answer label "A/B/C/D" and not a direct source-index code: it sends the first B bytes of the source-selected public candidate string and decodes by lexical overlap. Explicit source-index in Table 3 means the source label alone, without target calibration; richer source/target pair rules are treated as diagnostics. For 4-way multiple choice, the selected-candidate boundary is only 2 bits information-theoretically, although our implementation reports it as a one-byte label channel. This matters because a packet can help while mostly preserving the source's selected candidate; that is no-text candidate transfer, not proof of a shared latent language or compression beyond selected-candidate communication.

Threat model. The packet channel hides source text and dense source state, but it is not identity-hiding, choice-hiding, or formally private in the differential-privacy, cryptographic, or information-theoretic secrecy sense. The sender may privately compute over the public prompt and choices, then transmit only a bounded byte string. The receiver may use that byte string, the public prompt, and its own computation. It may not observe source generations, chain-of-thought, hidden states, KV cache, logits, score vectors, or labels. A passive packet observer sees only packet bytes and framing, not source internals or text; however, the current packet leaks the source's preferred candidate with 0.995 mean match on ARC-Challenge and 0.999 on OpenBookQA. A method fails the workshop claim if its apparent gain is matched by target-only effects, same-budget visible text, explicit source-index/rank/score communication, wrong-row transfer, or answer leakage.

Control suite. Table 1 lists the reviewer-facing controls used to interpret the reported rows. The suite is intentionally stronger than the headline claim: several controls are designed to collapse methods that only copy the source's preferred answer choice.

3 Public-coordinate packets

The current ARC-Challenge implementation uses a public train-anchor chart. For each candidate, we compute a public relative-coordinate feature against a frozen set of training anchors, then project that feature into a low-frequency orthonormal DCT-II spectral

Control family	What it tests	Current role
target-only / zero-source	target-cache or no-packet effects	required baseline
same-budget visible text	whether text at the same byte budget suffices	required baseline
source-index / rank / score	selected-candidate shortcuts	main claim boundary
wrong-row / row shuffle	row-specific source causality	destructive control
same-source-choice wrong-row	source-choice-only transfer	hardening control
anchor/bin/coordinate shuffle	shared-coordinate dependence	mechanism control
candidate roll/derangement	answer-order artifacts	destructive control
source-family substitution	cross-family generality	falsification control
cached connector repair	easy learned receiver rescue	negative diagnostic

Table 1: Control suite. The workshop claim requires packet utility to be interpreted through these controls; the current positive is explicitly bounded by source-index and source-family failures.

Step	Paper implementation
Source evidence	The source privately selects a candidate from the public choices using answer-key-forbidden source caches; no source text, logits, hidden states, KV cache, or labels cross the interface.
Public chart	Build candidate-pair text anchors from train rows; ARC-Challenge uses 384 public anchors and a 96-dimensional low-frequency DCT-II basis.
Source coordinates	Compute source candidate residuals in the public chart, project into the DCT basis, and keep the largest signed coordinates.
Packet	Encode each selected coordinate as index, sign, and quantized magnitude; ARC-Challenge uses 4 nonzero coordinates in 8 raw bytes / 11 framed bytes, and OpenBookQA uses 1 coordinate in 3 raw bytes / 6 framed bytes.
Receiver decode	Compute receiver-side candidate residuals in the same public basis, normalize candidate codes, and pick the candidate with highest cosine compatibility to the decoded packet vector.
Validation	Packet budgets and basis settings are selected before frozen test readout; source-index, label-text, rank-code, random-index, text, wrong-row, and shuffle controls are evaluated on the same frozen rows.

Table 2: Algorithm sketch for the reported packet rows. The table is intentionally explicit because the workshop claim depends on separating the no-text packet from direct source-index or source-label transmission.

basis. The packet is a sparse signed syndrome in this shared coordinate system. The receiver/decoder scores candidate compatibility with the packet in the same public basis, combining the packet evidence with receiver-side candidate information.

Concretely, let $\phi_i(x) \in \mathbb{R}^d$ be the public anchor feature for candidate c_i, and let

$$r_i = \phi_i(x) - \frac{1}{m}\sum_{\ell=1}^{m} \phi_\ell(x), \qquad u_i = U_k^\top r_i,$$

where U_k is the low-frequency DCT-II basis. If the source selects candidate s, the encoder keeps only the largest signed coordinates,

$$z = \{(j, \mathrm{sign}(u_{s,j}), Q(|u_{s,j}|)) : j \in \mathrm{TopK}(|u_s|)\}.$$

The decoder unpacks z into a sparse vector $\tilde{u}(z)$ and predicts

$$\hat{y} = \arg\max_i \cos\left(\tilde{u}(z), u_i^{(R)}\right),$$

with validation-selected packet budget and basis settings fixed before frozen test readout.

The design is related to relative representations (Moschella et al., 2023): anchors provide a coordinate system less tied to raw hidden dimensions. Our use is narrower. We do not claim zero-shot latent stitching or global representation alignment; we use a public basis to define a tiny task packet. Random shared anchors work nearly as well as semantic anchors in the current hashed ARC-Challenge implementation, so the correct interpretation is shared coordinate agreement, not semantic anchor names.

Benchmark	split	bytes	seeds	src	packet	target	text	pkt-src	pkt-text
ARC-Challenge	test, 1172	8/11	10	0.346	0.344	0.265	0.300	-0.008	+0.025
OpenBookQA	test, 500	3/6	20	0.378	0.378	0.276	0.350	-0.006	-0.002

Table 3: Main audit rows. "Bytes" reports payload/framed bytes. "Src" is explicit answer-key-forbidden source-index accuracy before packet decoding. "Text" is the same-budget structured text control. "Pkt-src" and "pkt-text" are minimum paired-bootstrap 95% lower bounds for packet minus source-index and packet minus same-budget text. Packet-minus-target lower bounds are +0.038 on ARC-Challenge and +0.046 on OpenBookQA.

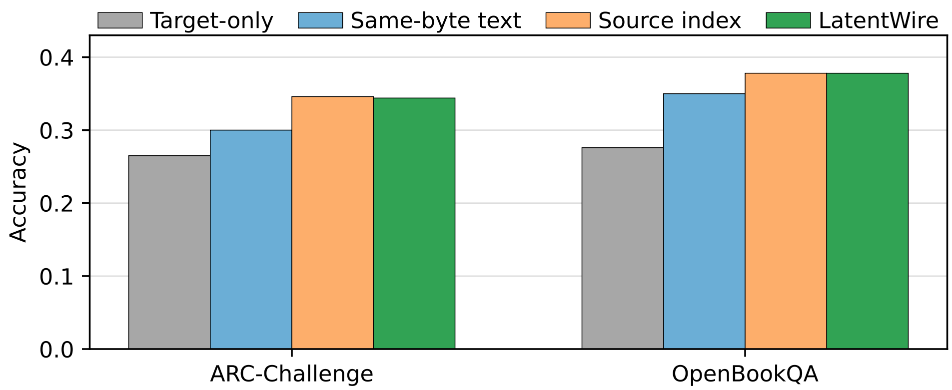

Figure 2: Packet accuracy against target-only, same-budget structured text, and explicit source-index controls. The source-index bar is the decisive boundary: it matches or exceeds the packet on both headline rows.

OpenBookQA uses the same packet-evaluation contract at a smaller 3-byte budget. Across both tasks, the source-choice cache is answer-key-forbidden: the source sees public questions and choices, not labels. The headline source cache is from `Qwen/Qwen2.5-0.5B-Instruct` scored on CPU with mean-normalized choice log-likelihoods; the packet receiver is the public hashed/DCT candidate decoder rather than a live target-model forward. The validation-only score-sketch diagnostic separately uses `Qwen/Qwen3-0.6B` receiver scores. The source-family falsification uses `microsoft/Phi-3-mini-4k-instruct`, and the cached connector diagnostic uses `TinyLlama/TinyLlama-1.1B-Chat-v1.0`. The packet is evaluated on frozen test slices after validation selection.

4 Main results

Table 3 and Figure 2 summarize the current positive evidence and the source-index audit. The ARC-Challenge row uses 10 packet/control seeds and 2000 paired bootstrap samples per seed on 1172 test examples. The OpenBookQA row now uses 20 packet/control seeds on the 500-example test slice. The source-index audit is generated by reconstructing the same packet rows and adding explicit source-index, source-label-text, source-rank-code, and entropy-matched random-index baselines.

Reproducibility card. All headline rows are frozen test-slice evaluations. The ARC-Challenge row uses the Qwen2.5 source-choice cache, 10 seeds, and 2000 bootstrap samples; the OpenBookQA row uses the same no-source-text contract with 20 projection/control seeds and 1000 bootstrap samples; the source-index audit reruns the same frozen prediction surfaces with explicit index/rank/label baselines. Exact input paths, hashes, scripts, and rerun status are listed in the artifact manifest and regenerated by the COLM_v3 review-packet script.

On ARC-Challenge, the matched Fourier/anchor-syndrome packet reaches 0.344 mean test accuracy with minimum seed accuracy 0.342. The answer-key-forbidden source index itself reaches 0.346, and packet predictions match the source-selected candidate at 0.995 mean rate across seeds. Anchor-ID shuffle, anchor-value shuffle, and spectral-bin permutation controls all fail in 0/10 seeds and collapse near target-only. The random shared-anchor diagnostic passes, which weakens any claim that named anchors are semantically special, but strengthens the claim that a shared public coordinate chart is the relevant mechanism.

On OpenBookQA, the 3-byte packet reaches 0.376–0.380 across 20 seeds. The explicit source-index baseline reaches 0.378, and the packet follows the source index at 0.999 mean rate. The largest candidate-derangement control accuracy is 0.212. Although the packet is +0.028 above same-budget structured text in mean, the expanded paired-bootstrap lower bound is -0.002, so this is not a statistically supported text-baseline improvement. This second task reduces the risk that the result is only an ARC-Challenge artifact, but it confirms the same boundary: the current method is source-choice preserving and same-family, not cross-family latent reasoning.

The practical readout is an evidence ladder rather than a single win/loss row. The packets beat target-only on both headline rows; ARC-Challenge clears same-budget text, while OpenBookQA does not clear the paired lower-bound gate against text. Coordinate and candidate-destructive controls prevent treating the result as random packet regularization; explicit source-index/rank controls prevent overstating the mechanism. In framed-byte terms, the target-only lift is $0.079/11 = 0.0072$ accuracy per byte on ARC-Challenge and $0.102/6 = 0.0170$ accuracy per byte on OpenBookQA. Those rates are useful for comparing packet regimes, but they are not evidence that the current packet is better than a direct selected-candidate code.

Claim boundary. The current packet does not improve over explicit source-index communication: source-index reaches 0.346 on ARC-Challenge and 0.378 on OpenBookQA, while the packet reaches 0.344 and 0.378. A two-stage seed/item cluster bootstrap for packet minus source-index gives ARC-Challenge mean -0.0016 with 95% CI $[-0.0030, -0.0003]$ and OpenBookQA mean -0.0001 with 95% CI $[-0.0009, +0.0007]$ after expanding to 20 seeds. We therefore claim no-text fixed-byte candidate transfer under destructive controls, not compression beyond a selected-candidate channel.

Disagreement audit. A source/target/packet disagreement audit makes the mechanism visible. On ARC-Challenge, source-only-correct rows account for 0.265 of seed-items and target-only-correct rows for 0.185; the packet follows the source in 0.995 of seed-items, repairs target errors in 0.264, and damages target-correct rows in 0.185. On OpenBookQA, the corresponding rates are 0.266, 0.164, 0.999, 0.267, and 0.164. This is useful transfer when the source is right and the target is wrong, but it also confirms that the packet is mostly source-choice transport.

Score-quantization diagnostic. The frozen headline ARC-Challenge/OpenBookQA source-choice caches store top choices, not raw calibrated score vectors, so source-score quantization is not a full headline baseline. We include a validation-only ARC-Challenge diagnostic with a 3-byte quantized source score distribution and receiver score side information. It reaches the same held-out accuracy as always using the source top label (0.389) and does not beat the top-label pair rule. This supports the same boundary: simple score sketches do not yet demonstrate information beyond source-choice transfer.

Breadth audits. Additional benchmark/model rows remain reviewer-packet diagnostics, not headline evidence. They include HellaSwag validation-1024 (Zellers et al., 2019), CommonsenseQA (Talmor et al., 2019), SciQ (Welbl et al., 2017), and a latest-model emitter audit. These rows improve visibility into breadth and failure modes, but they do not strengthen the main ARC-Challenge/OpenBookQA claim.

Benchmark	seeds	pkt-target	pkt-text	pkt-source	provenance
ARC-Challenge	10	+0.038	+0.025	-0.008	frozen test, hash-audited
OpenBookQA	20	+0.046	-0.002	-0.006	frozen test, hash-audited

Table 4: Uncertainty summary for the main audit rows. Values are paired-bootstrap 95% lower bounds for packet accuracy minus the named comparator. The negative lower bound against same-budget text on OpenBookQA is intentionally visible: that row is a narrow secondary positive, not a text-baseline separation.

Source	surface	n	packet	target	text	ref. pkt	blocker
Phi-3	full test	1172	0.244	0.265	0.232	–	source quality
Phi-3	Qwen-disagree	833	0.200	0.273	0.203	0.340	source quality
TinyLlama	Qwen-disagree	473	0.269	0.268	0.258	0.317	family mismatch

Table 5: Cross-family diagnostics. "Ref. pkt" is the Qwen-substituted packet on disagreement rows. Validation-incomplete stronger-Qwen rows are kept out of the main paper and remain reviewer-packet diagnostics only.

5 Falsification and failure decomposition

The most important negative result is the Phi-3 source-family replacement. We keep the same ARC-Challenge packet budget, public basis, seed count, and bootstrap budget, but replace the Qwen source-choice cache with a Phi-3 source. The packet no longer passes. On the full ARC-Challenge test slice, the Phi-3 packet reaches 0.244 versus target-only 0.265. On the stricter Qwen-disagreement slice, where the alternate source and Qwen source disagree, the Phi-3 packet reaches 0.200 versus 0.340 for the Qwen-substituted packet. A new failure diagnosis shows why this row fails at the observed interface: Phi-3's source-choice accuracy is 0.246 versus Qwen's 0.346 on the same test cache, Phi-3 and Qwen choose the same candidate only 0.289 of the time, and the decoded packet follows the Phi-3 source choice at 0.997. The packet is therefore transporting the alternate source's weaker candidate preference rather than corrupting the packet during decoding.

The failure decomposition separates three causes: source endpoint quality, packet corruption, and common-feature mismatch. For Phi-3, packet predictions follow the Phi-3 source choice at 0.997, but the source itself is below the target and below the same-family Qwen source. Conditional rows sharpen the diagnosis: when Phi-3 alone is correct, the Phi-3 packet is correct in 0.997 of seed-items; when Qwen alone is correct, the Phi-3 packet is correct in 0.000. For TinyLlama, packet fidelity is also high, but the source-family mismatch causes the disagreement-slice packet to lose to Qwen-substituted packets. Across these rows, the packet mostly transports the source-selected candidate through the public coordinate basis. The source-index audit in Table 3 closes one obvious reviewer question: the current packet does not beat a direct selected-candidate code. The next research gate is therefore a richer common-feature connector or calibrated source-score packet that can improve beyond packet-only source choice.

We also tested cached learned repairs on the available local artifacts. A candidate-syndrome connector that sees packet and score-shape features reaches 0.288 versus 0.317 for Qwen-substituted packets on frozen TinyLlama/Qwen disagreement rows. A one-hidden-layer MLP over cached TinyLlama hidden/query summaries reaches 0.232, below Qwen-substituted packets, cached Tiny packets, target-only, and same-budget text. These negatives are useful: simple cached score geometry and mean hidden/query summaries do not explain the remaining oracle headroom.

6 Systems boundary

The systems contribution is byte/exposure accounting, not native serving throughput. Table 6 compares framed packet bytes with one-token KV/cache-state accounting floors

Method	raw	framed	text	KV/hidden	claim status
LatentWire OpenBookQA packet	3B	6B	no	no	local positive
LatentWire ARC-Challenge packet	8B	11B	no	no	local positive
Same-byte text control	8B	11B	yes	no	control only
1-bit/KV-element floor	768B	768B	no	KV	floor only
KIVI 2-bit KV floor	1536B	1536B	no	KV	floor only
C2C one-token KV floor	12288B	12288B	no	KV	native pending

Table 6: Exposure classes. Packet rows are measured/accounted local communication objects. KV/cache rows are analytical byte floors or native-pending baselines. The paper-ready systems claim is object size and source-exposure accounting, not GPU throughput, TTFT, TPOT, HBM, energy, or C2C superiority.

from neighboring systems and quantization methods. LatentWire sends 6–11 framed bytes on the two headline rows. A one-token 1-bit-per-KV-element accounting floor is 768 bytes in our model-configuration accounting; the largest headline packet is therefore 69.8x smaller as a message object than that floor. This is not a function-equivalence claim. C2C-style cache transfer, KVComm/KVCOMM-style selective sharing, and serving substrates such as vLLM/PagedAttention and SGLang operate on KV/cache objects rather than shape-restricted task hints (Fu et al., 2026; Shi et al., 2026; Ye et al., 2025; Kwon et al., 2023; Zheng et al., 2024). KV quantization work reduces cache bytes but still communicates or stores model state (Zandieh et al., 2024; Liu et al., 2024; Hooper et al., 2024; Zandieh et al., 2026). We also ran a local dense-baseline smoke gate: the published C2C artifact executes on converted ARC/OpenBookQA letter-generation prompts on MPS, while the current KVComm port still fails at Qwen3 cache/mask compatibility after eager-attention and DynamicCache shims. These smoke rows are runtime-readiness evidence only, not matched headline baselines.

We label packet-object bytes as measured/accounted local artifacts and dense-state rows as analytical or literature-derived byte floors. This distinction is important: the current data support privacy/exposure and byte-budget comparisons, but not claims about GPU latency, HBM traffic, energy, or serving throughput.

7 Related work

Latent and relative representation communication. Relative representations use similarities to anchors to enable latent-space communication under invariances (Moschella et al., 2023). Sparse autoencoder work motivates interpretable feature coordinates (Huben et al., 2024), while feature-space universality work suggests that some features can be related across models (Lan et al., 2024). Crosscoders, transcoders, and activation-transfer methods are natural future sources of interpretable packet atoms or dense communication baselines. SAEBench provides evaluation machinery for sparse autoencoders (Karvonen et al., 2025), but these works do not by themselves provide a byte-counted packet protocol or task-level evidence transfer result under source-choice controls.

Embedding communication and steganography. CIPHER-style embedding communication and embedding-space steganography are closer in spirit than text prompting because they communicate through non-natural-language vectors or geometry (Pham et al., 2024; Westphal et al., 2026). LatentWire differs in bandwidth and interface: the communicated object is a discrete packet decoded through a public chart, not a continuous embedding stream inserted into a model. The steganography literature is also a reminder that "private" terminology should be quantitative; here we make only a content/state exposure claim and explicitly report source-choice leakage.

Destructive controls. Our control suite is comprehensive, but it is not novel as a methodology in kind. It borrows from control tasks, saliency sanity checks, adversarial/heuristic dataset controls, and behavioral tests (Hewitt & Liang, 2019; Adebayo et al., 2018; McCoy et al., 2019; Ribeiro et al., 2020). The paper's claim is the application and aggregation of

Regime	Example	Object crossing interface	Difference from LatentWire
dense cache fusion	C2C	projected source KV cache	high-bandwidth, source-state exposing
selective KV sharing	KV-cache comm.	selected KV pairs/layers	cache-level, native pending
KV compression	KV quantizers	compressed KV/cache state	shrinks dense state, not task packets
serving substrate	vLLM/SGLang	KV reuse and memory management	runtime baseline, not communication protocol
prompt/prefix compression	prefix/gist tokens	prompt-like tokens or parameters	not no-text model-to-model evidence
local shortcut baselines	source-index/rank/score	selected answer or score sketch	must be beaten for stronger claims

Table 7: Related-work and baseline matrix. LatentWire is positioned as low-byte no-text packet transfer with destructive controls, not as a replacement for dense cache fusion or serving-runtime optimization.

those controls for byte-scale inter-model packets, not invention of falsification as a new evaluation paradigm.

Prompt compression and learned prefixes. Prefix tuning and gist tokens learn continuous or discrete prompt-like interfaces within language models (Li & Liang, 2021; Mu et al., 2023); text-compression systems such as LLMLingua and LongLLMLingua are stronger visible-text baselines for future work (Jiang et al., 2023a;b). LatentWire differs because the communicated object is an opaque byte packet decoded by a public receiver, not a learned soft prompt injected into one target or a compressed visible prompt. Query-bottleneck connectors such as BLIP-2's Q-Former show that learned bottlenecks can bridge frozen model families across modalities (Li et al., 2023); our failed cached hidden/query summaries suggest that a true tokenwise query-bottleneck is the right future branch, but it is not closed here.

Cache communication and systems. Cache-to-cache, KVComm, and KVCOMM communicate richer internal state and are the closest competitors for model-to-model communication (Fu et al., 2026; Shi et al., 2026; Ye et al., 2025). vLLM and SGLang improve serving through KV-cache management and reuse (Kwon et al., 2023; Zheng et al., 2024). KV quantizers such as QJL, KIVI, KVQuant, and TurboQuant reduce state cost but remain state-transfer or state-storage techniques (Zandieh et al., 2024; Liu et al., 2024; Hooper et al., 2024; Zandieh et al., 2026). Table 7 summarizes how these baselines differ from LatentWire. Our current systems result is a boundary artifact against these state objects, not a native throughput win.

8 Limitations and next gate

The main limitation is scope. The positive results are same-family/source-cache results on public multiple-choice tasks, and the packet largely preserves the source's selected candidate. We therefore do not claim to beat explicit source-index communication, source-score quantization, or calibrated source-choice baselines. We expanded OpenBookQA from 5 to 20 packet/control seeds; the result remains stable versus target-only but still does not clear same-budget text or source-index gates. We also attempted dense-baseline execution locally. C2C now has a same-task smoke path on converted ARC/OpenBookQA letter-generation prompts, but the smoke rows are too small and prompt-format-sensitive to serve as a headline baseline; KVComm reaches model loading and source-cache construction but still fails at Qwen3 cache/mask compatibility in the current local port. Finally, Phi-3 is no longer an unexplained falsification row at the packet interface: the packet follows Phi-3 faithfully, but Phi-3's source choices are weaker and often different from Qwen's. What remains unresolved is a deeper representational diagnosis and a positive cross-family connector. More importantly, 4-way multiple-choice tasks have a low answer-channel ceiling: a selected-candidate code needs only 2 bits information-theoretically and one byte in our implementation. A higher-entropy task, calibration/abstention metric, or multi-step downstream use is needed to demonstrate packet information beyond selected-candidate transport. TinyLlama common-feature repair fails, the validation-only score-sketch diagnostic does not beat source-label transfer, and a true cross-family learned query-bottleneck remains future work. We also do not claim formal Slepian–Wolf/Wyner–Ziv guarantees, formal privacy, or superiority over C2C, KVComm/KVCOMM, QJL, KIVI, TurboQuant, vLLM, or SGLang in native serving metrics.

The next exact gate for an ICLR-strength claim is a frozen-endpoint tokenwise connector with an explicit byte or query budget, tested against C2C/KV-style and structured-text baselines, with source-destroying controls and strict same-family versus cross-family separation. The current strongest claim is narrower: fixed-byte no-text packets can transfer useful same-family candidate evidence on ARC-Challenge and OpenBookQA, and the accompanying control ladder clarifies where the method stops working.

9 Conclusion

LatentWire shows that a tiny no-text candidate packet can improve receiver decisions on two public QA benchmarks under strict same-family controls. The same evidence draws a hard boundary: the current method is mostly source-choice preserving, is not a universal latent language, and does not yet solve cross-family communication. This combination of positive packet transfer, negative controls, and systems byte/exposure accounting is the paper's core.

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A Claim-boundary audit

Table 8 states the claim boundary used to write the main paper. The full generated packet is in `results/latentwire_colm_v3_review_packet_20260505/`.

Claim	Status	Evidence	Required wording
no-text packet protocol	supported	packet rows and controls	protocol/evaluation contribution
narrow same-family packet utility	supported but narrow	ARC-Challenge and OpenBookQA rows	source-candidate transfer
shortcut collapses are common	supported	source-index and wrong-row audits	claim boundary, not failure-only story
beats C2C or dense KV transfer	unsupported	no native matched C2C row	do not claim
matched C2C/KVComm comparison	partially attempted	C2C same-task smoke; KVComm cache-mask blocker	future scaled baseline
Phi-3 failure mechanism	diagnosed at interface	source-choice/family diagnostic	still no cross-family connector
OpenBookQA seed stability	improved	20 packet/control seeds	text/source-index gates still unresolved
native GPU/HBM/energy win	unsupported	no native NVIDIA measurement	future runbook only
broad latent/cross-family communication	unsupported	Phi-3 and connector failures	ICLR future goal

Table 8: Supported and unsupported claim boundaries. Each main-text claim maps to one supported or supported-but-narrow row.

B Artifact manifest

- ARC-Challenge headline: `colm_final/evidence/results/source_private_arc_challenge_fourier_anchor_syndrome_gate_20260502_budget8_10seed_b2000`.
- Phi-3 falsification: `colm_final/evidence/results/source_private_arc_challenge_source_family_cache_falsification_20260502_phi3_cpu_budget8_10seed_b2000`.
- Cross-family decomposition: `colm_final/evidence/results/source_private_arc_cross_family_failure_decomposition_20260502`.
- Systems boundary: `colm_final/evidence/results/source_private_systems_boundary_figure_table_20260502`.
- OpenBookQA headline: `results/source_private_openbookqa_seed_stability_20260505_qwen05_hashed_test_3b_20seed`.
- Source-index audit: `results/source_private_colm_acceptance_baselines_20260505_obqa20`.
- Aggregate source-index CI: `results/source_private_colm_acceptance_baselines_20260505_obqa20/aggregate_source_index_ci.md`.
- Phi-3 failure diagnosis: `results/latentwire_phi3_failure_diagnostic_20260505`.
- Dense baseline smoke: `paper/latentwire_dense_baseline_smoke_20260505.md`.
- Disagreement audit: `results/source_private_colm_acceptance_baselines_20260505_obqa20/disagreement_audit.md`.
- Score-sketch diagnostic: `results/source_private_arc_score_fusion_packet_probe_20260503_qwen05_qwen3_validation`.
- COLM_v3 review packet: `results/latentwire_colm_v3_review_packet_20260505`.

C Reproducibility summary

The artifact bundle records exact shell commands, input hashes, output hashes, and rerun results. The paper rows are produced from cached, answer-key-forbidden source-choice artifacts with the following settings:

Row	script	seeds	key settings
ARC-Challenge	Fourier/anchor syndrome gate	10	8B, 2000 bootstraps
OpenBookQA	seed-stability gate	20	3B, hashed features, 1000 bootstraps
Source-index audit	acceptance baseline audit	10/20	2000 bootstraps, 2/3/4/8B rate curve
Aggregate pkt-src CI	aggregate CI script	10/20	10000 two-stage seed/item bootstraps
Phi-3 diagnosis	failure diagnostic script	10	source-choice/family decomposition
C2C/KVComm smoke	dense MCQA generation smoke	–	MPS C2C smoke; KVComm cache-mask blocker
Phi-3	source-family cache falsification	10	8B, cached Phi-3 source
Systems	systems boundary table	–	byte/exposure accounting only

Headline artifacts are hash-verified frozen outputs, and rerun scripts are provided. Exact byte-identical reruns may differ only in timestamped JSON fields or local timing measurements; metric rows are regenerated from the tracked prediction artifacts.

D Per-seed audit rows

Table 9 exposes the per-seed test rows behind the main source-index audit. The important reviewer readout is sign consistency: packet-minus-source-index is never a positive lower-bound gate on either benchmark, while ARC-Challenge consistently clears same-budget text and OpenBookQA does not.

Benchmark	seed	packet	source-index	text	follows source	CI pkt-source
ARC-Challenge	47	0.342	0.346	0.300	0.997	-0.007
ARC-Challenge	53	0.346	0.346	0.300	0.995	-0.004
ARC-Challenge	59	0.343	0.346	0.300	0.990	-0.008
ARC-Challenge	61	0.345	0.346	0.300	0.997	-0.003
ARC-Challenge	67	0.343	0.346	0.300	0.995	-0.007
ARC-Challenge	71	0.344	0.346	0.300	0.995	-0.005
ARC-Challenge	73	0.344	0.346	0.300	0.997	-0.005
ARC-Challenge	79	0.343	0.346	0.300	0.997	-0.006
ARC-Challenge	83	0.346	0.346	0.300	0.996	-0.003
ARC-Challenge	89	0.345	0.346	0.300	0.995	-0.005
OpenBookQA	47	0.378	0.378	0.350	1.000	0.000
OpenBookQA	53	0.378	0.378	0.350	0.996	-0.006
OpenBookQA	59	0.378	0.378	0.350	1.000	0.000
OpenBookQA	61	0.380	0.378	0.350	0.998	0.000
OpenBookQA	67	0.378	0.378	0.350	1.000	0.000
OpenBookQA	71	0.378	0.378	0.350	1.000	0.000
OpenBookQA	73	0.376	0.378	0.350	0.998	-0.006
OpenBookQA	79	0.378	0.378	0.350	1.000	0.000
OpenBookQA	83	0.378	0.378	0.350	0.996	-0.006
OpenBookQA	89	0.378	0.378	0.350	1.000	0.000
OpenBookQA	97	0.378	0.378	0.350	1.000	0.000
OpenBookQA	101	0.378	0.378	0.350	1.000	0.000
OpenBookQA	103	0.380	0.378	0.350	0.998	0.000
OpenBookQA	107	0.376	0.378	0.350	0.996	-0.006
OpenBookQA	109	0.378	0.378	0.350	1.000	0.000
OpenBookQA	113	0.378	0.378	0.350	0.998	0.000
OpenBookQA	127	0.378	0.378	0.350	0.998	0.000
OpenBookQA	131	0.380	0.378	0.350	0.998	0.000
OpenBookQA	137	0.376	0.378	0.350	0.998	-0.006
OpenBookQA	139	0.376	0.378	0.350	0.998	-0.006

Table 9: Per-seed test audit. "CI pkt-source" is the per-seed paired-bootstrap 95% lower bound for packet minus explicit source-index. Full CSV is stored in `results/source_private_colm_acceptance_baselines_20260505_obqa20/per_seed_baseline_metrics.csv`.

E Artifact regeneration

The repository command

```
./venv_arm64/bin/python scripts/build_latentwire_colm_v3_review_packet.py
```

regenerates the claim audit, table/figure inventory, systems ledger, artifact manifest, and NVIDIA native runbook from tracked source artifacts. The disagreement audit is regenerated with

```
./venv_arm64/bin/python scripts/build_latentwire_colm_v3_disagreement_audit.py
```

F Lay explanation of the main experiment

The experiment asks whether a source model can send a tiny no-text hint to a public receiver/decoder that also sees the same question and answer choices. The sender chooses a compressed coordinate hint, only a few bytes long. The decoder uses that hint to choose an answer. If both sides agree on the coordinate system, the hint helps. If we scramble the coordinate system, the hint stops helping. That is evidence for a real shared packet interface in the tested same-family source-cache-to-public-decoder setting, while the failed Phi-3 run shows the interface is not yet general across model families.